

IPL Data Analytics

Initial Project Planning

1. Requirement Collection

Functional Requirements

User / Analyst Requirements

- Upload IPL datasets (match-level and ball-by-ball CSV files) for analysis.
- Automatic data cleaning and preprocessing of raw datasets.
- View team and player statistics (runs, wickets, strike rate, win rate).
- Filter statistics and visualizations by season, team, and venue.
- Analyze the impact of toss decisions on match outcomes.
- View win-loss patterns and season-wise team performance comparisons.
- View run and wicket distribution analysis across matches.
- Generate interactive visualizations (bar charts, line graphs, heatmaps, pie charts).
- View a consolidated analytical dashboard with key performance indicators (KPIs).
- Export or download analytical reports and charts.

Admin Requirements

- Secure admin login.
- Manage and update the underlying IPL dataset / database.
- Monitor overall system usage and data processing status.
- View logs of data uploads and dashboard usage.

Non-Functional Requirements

- User-friendly, intuitive interface.
- Fast query response and chart rendering time.
- Reliable and normalized data storage (MySQL).
- Scalable architecture to accommodate new seasons and datasets.
- Responsive dashboard design across devices.
- Maintainable, well-documented codebase.
- Secure handling of stored data.

2. Initial User Stories

US-01: As an analyst, I want to upload IPL datasets so that the system has data to analyze.

US-02: As a user, I want raw data to be automatically cleaned and preprocessed so that analysis is accurate.

US-03: As a user, I want to view team and player statistics so that I can assess performance.

US-04: As a user, I want to filter data by season, team, and venue so that I can focus on specific trends.

US-05: As a user, I want to analyze toss impact so that I can understand its influence on match outcomes.

US-06: As a user, I want to view win-loss patterns across seasons so that I can track team performance over time.

US-07: As a user, I want interactive charts for run and wicket distribution so that I can explore the data visually.

US-08: As a user, I want a summary dashboard with KPIs so that I can get insights at a glance.

US-09: As a user, I want to export analytical reports so that I can share or archive my findings.

US-10: As an administrator, I want to manage the database and monitor usage so that the system stays reliable and up to date.

3. Initial Database Schema

Teams

Field	Type / Key
team_id (PK)	INT
team_name	VARCHAR
short_code	VARCHAR
home_venue_id (FK)	INT

Matches

Field	Type / Key
match_id (PK)	INT
season	INT
team1_id (FK)	INT
team2_id (FK)	INT
toss_winner_id (FK)	INT
toss_decision	VARCHAR
venue_id (FK)	INT
winner_id (FK)	INT

Players

Field	Type / Key
player_id (PK)	INT
player_name	VARCHAR
team_id (FK)	INT
role	VARCHAR

Venues

Field	Type / Key
venue_id (PK)	INT
venue_name	VARCHAR

Field	Type / Key
city	VARCHAR

Deliveries

Field	Type / Key
delivery_id (PK)	INT
match_id (FK)	INT
inning	INT
over	INT
ball	INT
batsman_id (FK)	INT
bowler_id (FK)	INT
runs	INT
is_wicket	BOOLEAN

Design note: the schema is normalized (3NF). Matches and Deliveries form the analytical core, joined with Teams, Players, and Venues via foreign keys to support season, toss, and venue-based aggregation queries used by the dashboard.

4. Initial UI Sketches

Home Page

- IPL Data Analytics title / branding
- Upload Dataset button
- View Dashboard / Get Started button
- Brief project description

Analytics Dashboard

- Global filters: Season, Team, Venue
- KPI strip: Matches, Seasons, Venues Used, Players
- Team Win % by Season chart
- Toss Decision breakdown chart
- Venue Heatmap

Team / Player Stats Page

- Searchable stats table (runs, wickets, strike rate, win rate)
- Filter by season / team / venue
- Comparison charts between players or teams

Match & Toss Analysis Page

- Toss decision vs. match result comparison
- Run and wicket distribution charts
- Season-wise trend lines

Report Export Page

- Selected charts / tables preview
- Export as PDF / CSV options
- Download Report button

Admin Dashboard

- Total datasets uploaded
- System usage statistics
- Data upload / processing logs
- Database management controls